

# Thermodynamics

Lecture 3B:

- Property tables
- Ideal gas law

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# Cheat Sheet Available

Fall 2026

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## ME40 Thermodynamics Formula Sheet

### General Relationships

- Temperature conversion:  $T(K) = T(^{\circ}C) + 273.15$
- Extensive property:  $Z$
- Intensive property:  $z = \frac{Z}{m}$
- State postulate:  $z = z(x, y)$
- Definition of specific enthalpy:  $h = u + pv$
- Volumetric conversions:  $1 \text{ L} = 1000 \text{ dm}^3 = 0.001 \text{ m}^3$
- Interpolation:  $y = y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0}$
- Density and specific volume:  $v = \frac{1}{\rho}$

### Ideal Liquids

- Incompressible fluid:  $v = \text{const.}$
- Heat capacity of liquid:  $c_p(T) = c_v(T) = c(T)$

# After this lecture

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- Demonstrate the procedures for determining thermodynamic properties of pure substances.
- Understand equations of state for gases, and how and in what cases to apply them to specific thermodynamic problems.

# Learning objectives

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What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

# Learning objectives (II)

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- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

# Recap Lecture 3A

Saturation pressure/temperature

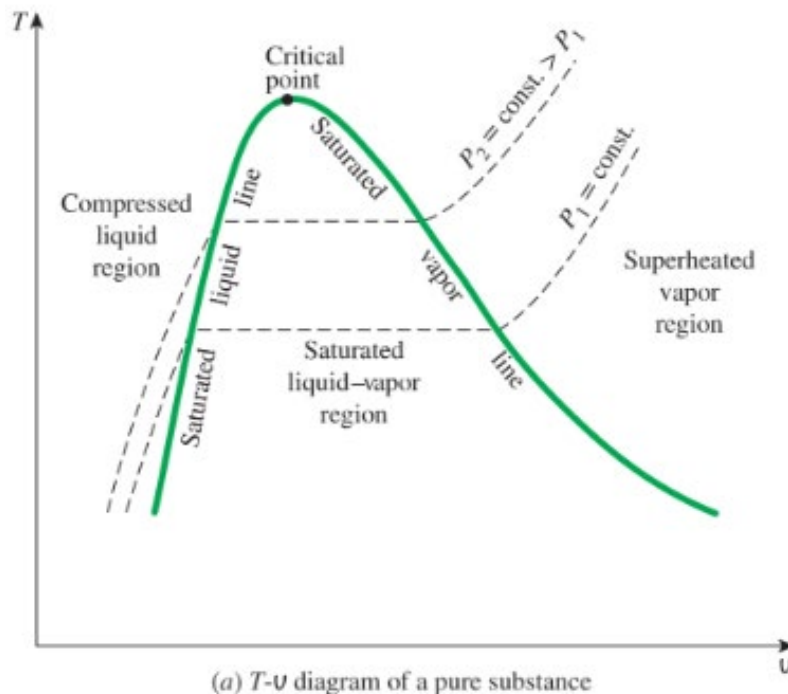
$$P_{sat} = f(T_{sat})$$

Vapor quality

$$x = \frac{m_g}{m_{total}}$$

Properties in the sat. liquid-vapor region

$$v_{avg} = (1 - x) v_f + x v_g$$

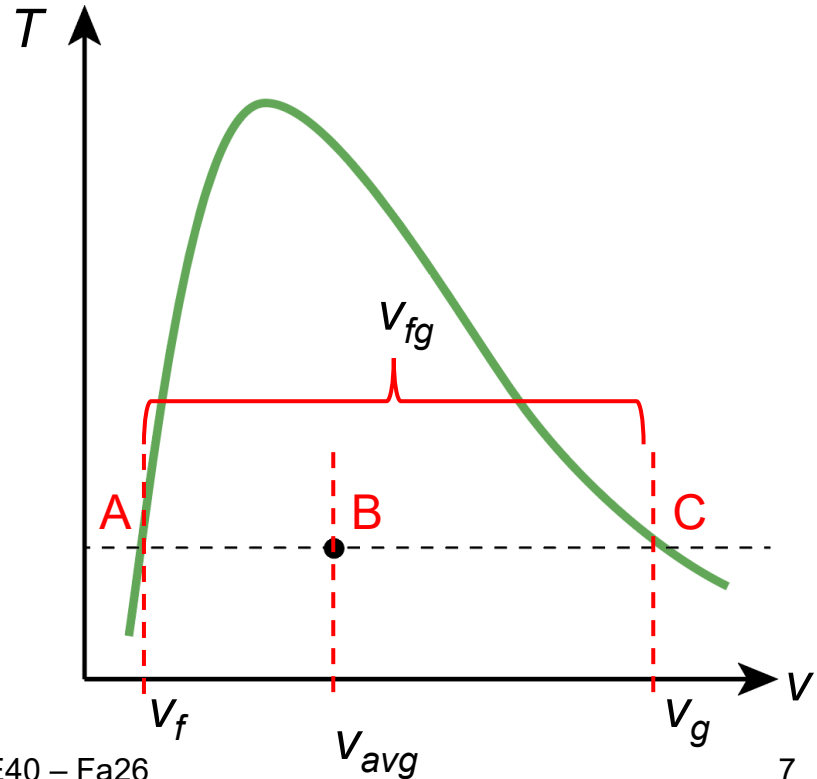


# Property tables: Saturated liquid vapor mixture

$$v_{avg} = v_f + x v_{fg}$$

$$x = \frac{v_{avg} - v_f}{v_{fg}}$$

$$x = \frac{m_{vapor}}{m_{total}} = \frac{\overline{AB}}{\overline{AC}}$$



# Superheated vapor

How to determine  $v(T = 50^\circ\text{C})$

$$v = v(T)$$

How to determine  $v(T = 75^\circ\text{C})$

$$v(T) = v_1 + (T - T_1) \frac{(v_2 - v_1)}{(T_2 - T_1)}$$

| $T$<br>°C                                       | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|-------------------------------------------------|---------------------------|--------------|--------------|----------------|
| $P = 0.01 \text{ MPa (} 45.81^\circ\text{C)}^*$ |                           |              |              |                |
| Sat. †                                          | 14.670                    | 2437.2       | 2583.9       | 8.1488         |
| 50                                              | 14.867                    | 2443.3       | 2592.0       | 8.1741         |
| 100                                             | 17.196                    | 2515.5       | 2687.5       | 8.4489         |
| 150                                             | 19.513                    | 2587.9       | 2783.0       | 8.6893         |
| 200                                             | 21.826                    | 2661.4       | 2879.6       | 8.9049         |
| 250                                             | 24.136                    | 2736.1       | 2977.5       | 9.1015         |
| 300                                             | 26.446                    | 2812.3       | 3076.7       | 9.2827         |

Excerpt from Superheated water table

# Compressed liquid water

- Approximation: in the absence of compressed liquid data, treat the liquid as saturated liquid at the given temperature.

$$v \approx v_f(T)$$

$$u \approx u_f(T)$$

$$h \approx h_f(T)$$

| $T$<br>°C                                  | $v$<br>m <sup>3</sup> /kg | $u$<br>kJ/kg | $h$<br>kJ/kg | $s$<br>kJ/kg·K |
|--------------------------------------------|---------------------------|--------------|--------------|----------------|
| $P = 5 \text{ MPa (263.94}^\circ\text{C)}$ |                           |              |              |                |
| Sat.                                       | 0.0012862                 | 1148.1       | 1154.5       | 2.9207         |
| 0                                          | 0.0009977                 | 0.04         | 5.03         | 0.0001         |
| 20                                         | 0.0009996                 | 83.61        | 88.61        | 0.2954         |
| 40                                         | 0.0010057                 | 166.92       | 171.95       | 0.5705         |
| 60                                         | 0.0010149                 | 250.29       | 255.36       | 0.8287         |
| 80                                         | 0.0010267                 | 333.82       | 338.96       | 1.0723         |
| 100                                        | 0.0010410                 | 417.65       | 422.85       | 1.3034         |

# Example 1: in groups of 2; discuss the following question

## Given:

You have a liquid-vapor mixture of water at  $p = 50 \text{ kPa}$  and  $x = 0.35$ .

## Question:

1. What is the temperature of the mixture in  $^{\circ}\text{C}$ ?
2. Determine the internal energy of the mixture in  $\text{kJ/kg}$ ?

## To Do:

- Extract the correct data from the table.
- Go to [pollev.com/claudiohail](https://pollev.com/claudiohail)
- Enter the first names of your group members, e.g. Olivia and Liam.
- ... and answer question 1.

| Press.,<br>$P \text{ kPa}$ | Sat.<br>temp.,<br>$T_{\text{sat}}$<br>$^{\circ}\text{C}$ | Specific volume,<br>$\text{m}^3/\text{kg}$ |                         | Internal energy,<br>$\text{kJ/kg}$ |                    |                         |
|----------------------------|----------------------------------------------------------|--------------------------------------------|-------------------------|------------------------------------|--------------------|-------------------------|
|                            |                                                          | Sat.<br>liquid,<br>$v_f$                   | Sat.<br>vapor,<br>$v_g$ | Sat.<br>liquid,<br>$u_f$           | Evap.,<br>$u_{fg}$ | Sat.<br>vapor,<br>$u_g$ |
| 25                         | 64.96                                                    | 0.001020                                   | 6.2034                  | 271.93                             | 2190.4             | 2462.4                  |
| 30                         | 69.09                                                    | 0.001022                                   | 5.2287                  | 289.24                             | 2178.5             | 2467.7                  |
| 40                         | 75.86                                                    | 0.001026                                   | 3.9933                  | 317.58                             | 2158.8             | 2476.3                  |
| 50                         | 81.32                                                    | 0.001030                                   | 3.2403                  | 340.49                             | 2142.7             | 2483.2                  |
| 75                         | 91.76                                                    | 0.001037                                   | 2.2172                  | 384.36                             | 2111.8             | 2496.1                  |



# Example 1 Solution

Given  $p = 50 \text{ kPa}$ ,  $x = 0.35$

Determine  $T$  and  $u$

$$T = T_{\text{sat}}(p) = 81.32 \text{ } ^\circ\text{C}$$

$$u = (1 - x) u_f + x u_g$$

$$u = 0.65 \cdot 340.49 + 0.35 \cdot 2483.2$$

$$u = 1090.438 \text{ kJ/kg}$$

| Press.,<br>$P \text{ kPa}$ | Sat.<br>temp.,<br>$T_{\text{sat}}$<br>$^\circ\text{C}$ | Specific volume,<br>$\text{m}^3/\text{kg}$ |                      | Internal energy,<br>$\text{kJ/kg}$ |                    |                      |
|----------------------------|--------------------------------------------------------|--------------------------------------------|----------------------|------------------------------------|--------------------|----------------------|
|                            |                                                        | Sat. liquid,<br>$u_f$                      | Sat. vapor,<br>$u_g$ | Sat. liquid,<br>$u_f$              | Evap.,<br>$u_{fg}$ | Sat. vapor,<br>$u_g$ |
| 25                         | 64.96                                                  | 0.001020                                   | 6.2034               | 271.93                             | 2190.4             | 2462.4               |
| 30                         | 69.09                                                  | 0.001022                                   | 5.2287               | 289.24                             | 2178.5             | 2467.7               |
| 40                         | 75.86                                                  | 0.001026                                   | 3.9933               | 317.58                             | 2158.8             | 2476.3               |
| 50                         | 81.32                                                  | 0.001030                                   | 3.2403               | 340.49                             | 2142.7             | 2483.2               |
| 75                         | 91.76                                                  | 0.001037                                   | 2.2172               | 384.36                             | 2111.8             | 2496.1               |



# Jupyter notebook example

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Click this link to launch the notebook in the Berkeley hub: [waterProperties Script](#)

## IAPWS Example

```
from iapws import IAPWS97
water = IAPWS97(P=p1_MPa, T=T1_K)
s1 = water.s # kJ/kg*K
h1 = water.h # kJ/kg
v1 = water.v # m3/kg
u1 = water.u # kJ/kg
```

## PyCalor Example

```
from pyCalor import thermo as th
st = th.state("water", p=(p1_MPa, "MPa"), t=(T1_K, "K"))
h = st.h # kJ/kg
s = st.s # kJ/(kg*K)
v = st.v # m^3/kg
```

# Example 2: in groups of 2; discuss the following question

## Given:

You have a superheated water vapor at  $T = 350^\circ\text{C}$  and 8 bar.

## Question:

1. Determine the specific volume of the the vapor in  $\text{m}^3/\text{kg}$ ?

## To Do:

- Extract the data from the table.
- Go to [pollev.com/claudiohail](https://pollev.com/claudiohail)
- Enter the first names of your group members, e.g. Olivia and Liam.
- ... and answer question 1.



|      | $v$<br>$\text{m}^3/\text{kg}$                  | $u$<br>$\text{kJ/kg}$ | $h$<br>$\text{kJ/kg}$ | $s$<br>$\text{kJ/kg}\cdot\text{K}$ |
|------|------------------------------------------------|-----------------------|-----------------------|------------------------------------|
|      | $P = 0.80 \text{ MPa } (170.41^\circ\text{C})$ |                       |                       |                                    |
| Sat. | 0.24035                                        | 2576.0                | 2768.3                | 6.6616                             |
| 200  | 0.26088                                        | 2631.1                | 2839.8                | 6.8177                             |
| 250  | 0.29321                                        | 2715.9                | 2950.4                | 7.0402                             |
| 300  | 0.32416                                        | 2797.5                | 3056.9                | 7.2345                             |
| 350  | 0.35442                                        | 2878.6                | 3162.2                | 7.4107                             |
| 400  | 0.38429                                        | 2960.2                | 3267.7                | 7.5735                             |
| 500  | 0.44332                                        | 3126.6                | 3481.3                | 7.8692                             |

# Lecture 1B: Thermodynamic state postulate

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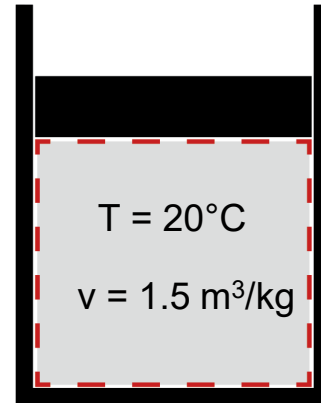
- Definition: A simple compressible system is completely defined by two independent intensive state properties.

$$T = f_1(p, v)$$

$$v = f_2(p, T)$$

$$p = f_3(v, T)$$

$$0 = f(T, p, v)$$



# Ideal gas law

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- For certain substances in the gas phase, we can use a simplified analytical description of their properties, and don't need to use thermodynamic tables.

$$pV = n\bar{R}T$$

$$pv = RT$$

$$pV = mRT$$

$$p = \rho RT$$

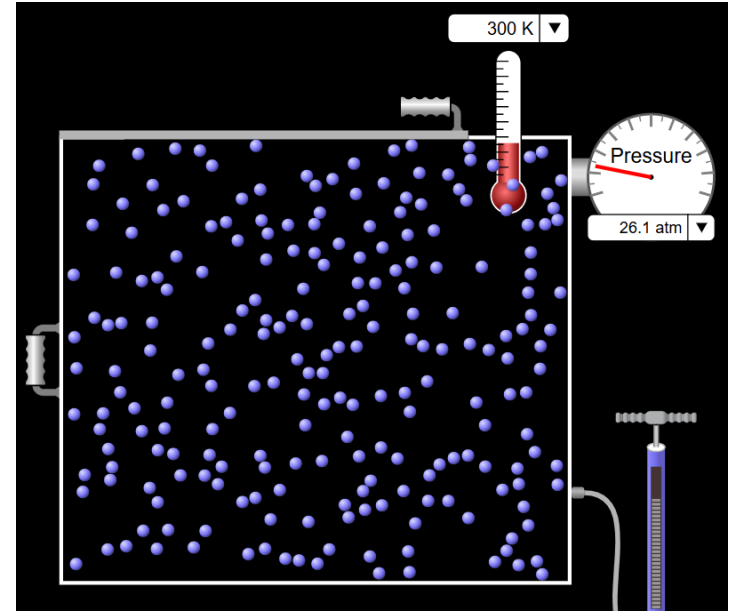
$$\bar{R} = 8.314 \frac{J}{molK}$$

$$R = \frac{\bar{R}}{M}$$

|                        |                       |
|------------------------|-----------------------|
| Pressure               | p [Pa]                |
| Volume                 | V [m <sup>3</sup> ]   |
| Temperature            | T [K]                 |
| Amount                 | n [mol]               |
| Universal gas constant | $\bar{R}$ [J/(K mol)] |
| Specific gas constant  | R [J/(K kg)]          |
| Molar mass             | M [kg/(mol)]          |

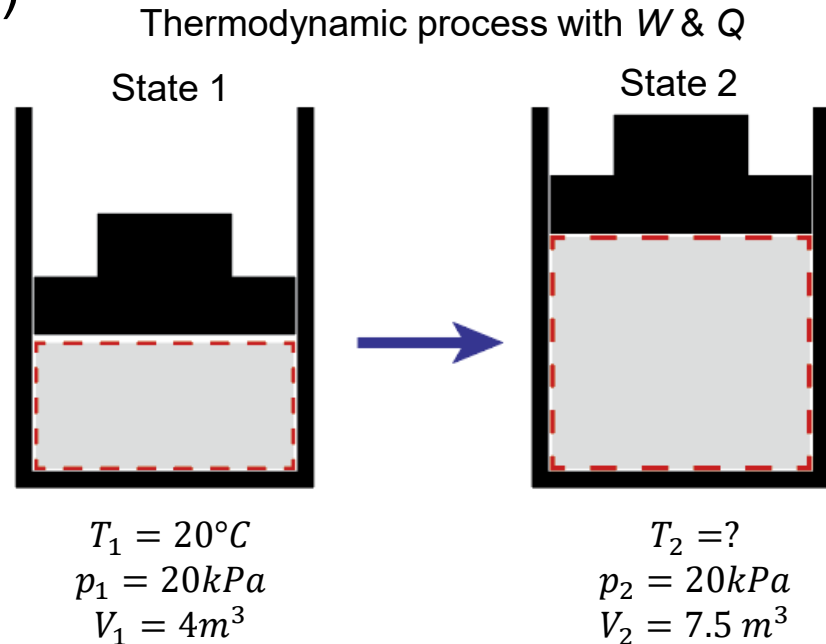
# What is an ideal gas?

- Gas molecules are constantly moving in random motion.
- Collision between gasses are perfectly elastic, i.e. there are no intermolecular forces.
- Gas molecules are of a negligible volume and behave like point masses.
- Good approximation for non-polar gasses at low pressures and high temperatures.
- Derivation from kinetic gas theory



# Example: ideal gas law

- An ideal gas ( $R = 287 \text{ J/kgK}$ ) is undergoing an expansion from state 1 to state 2  
Process on an ideal gas (see Figure)
- Determine the temperature of the gas in state 2.



# Example Solution: ideal gas law

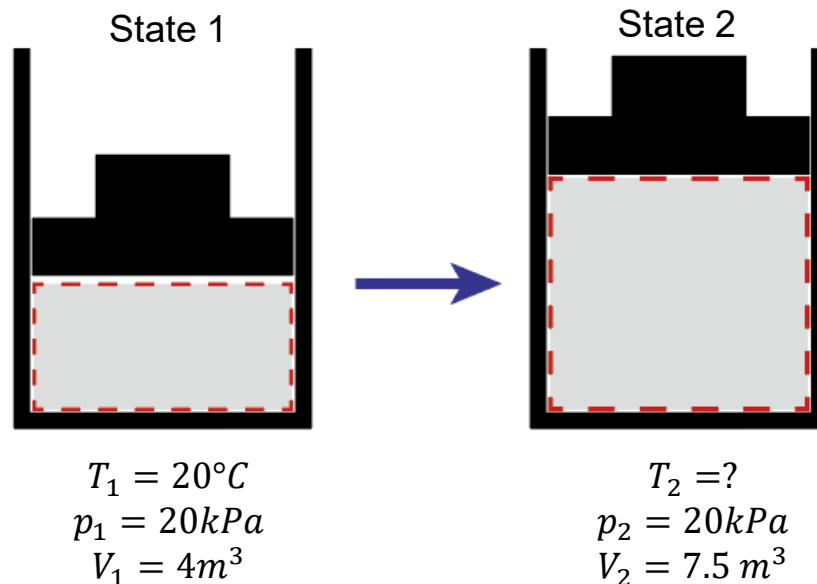
$$p_1 V_1 = mRT_1$$

$$m = \frac{pV}{RT} = 0.951 \text{ kg}$$

$$p_2 V_2 = mRT_2$$

$$T_2 = \frac{p_2 V_2}{mR} = 549.6 \text{ K}$$

Thermodynamic process with  $W$  &  $Q$



# Example Solution: ideal gas law

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General

$$\frac{pV}{T} = mR = \text{const.}$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

$$T_2 = \frac{p_2 V_2}{p_1 V_1} T_1$$

Isobaric process

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Isochoric process

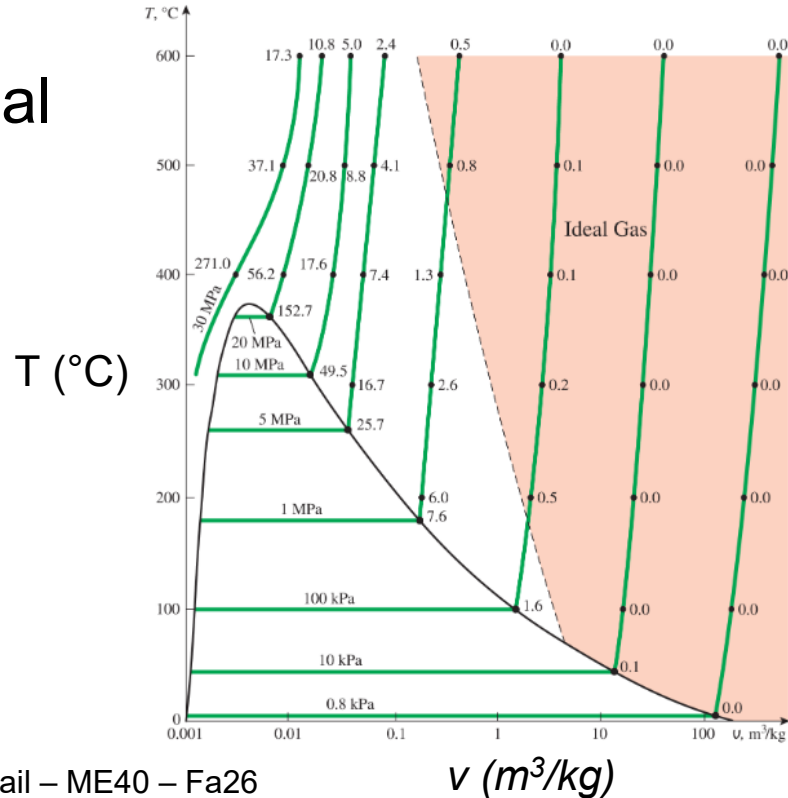
$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Isothermal process

$$p_1 V_1 = p_2 V_2$$

# Deviations from the ideal gas law (I)

- Percentage error in treating water as an ideal gas.



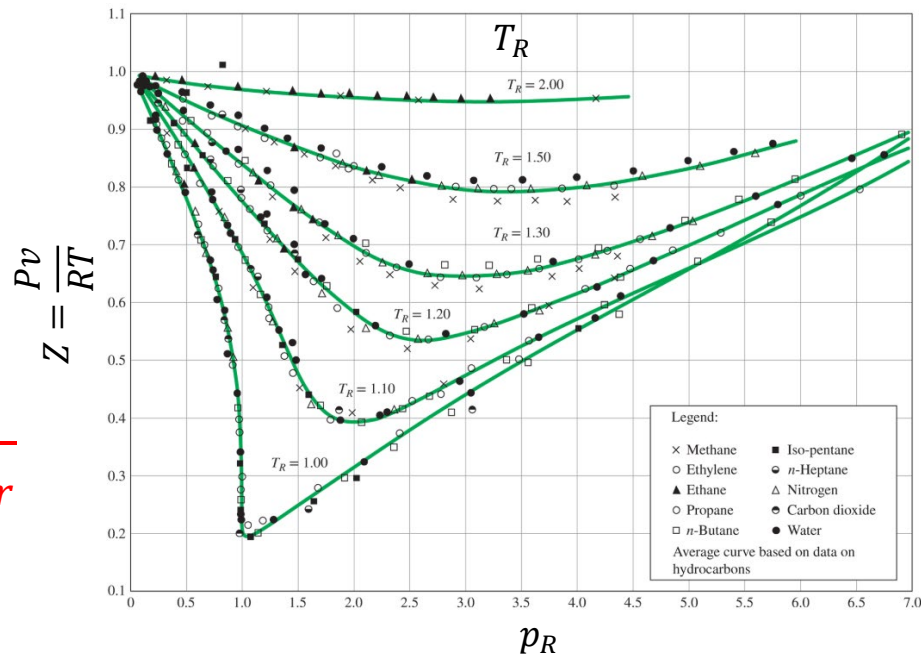
# Deviations from the ideal gas law (II)

- Tabulated compressibility factors

$$Z = \frac{Pv}{RT} = \frac{v_{actual}}{v_{ideal}}$$

Reduced properties definitions:

$$p_R = \frac{p}{p_{cr}}; T_R = \frac{T}{T_{cr}}; v_R = \frac{V}{RT_{cr}/p_{cr}}$$



# Other equations of state:

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Van der Waals Equation of State:

- Including intermolecular attraction forces.
- Volume occupied by molecules

$$\left(P + \frac{a}{v^2}\right)(v - b) = RT$$

Beattie-Bridgeman Equation of State:

- Using empirical constants

$$P = \frac{\bar{R}T}{\bar{v}^2} \left(1 - \frac{\epsilon}{\bar{v}T^3}\right) (\bar{v} + B) - \frac{A}{\bar{v}^2}$$

# Summary

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- Demonstrate the procedures for determining thermodynamic properties of pure substances.
- Understand equations of state for gases, and how and in what cases to apply them to specific thermodynamic problems.

# Next up

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- Energy analysis of closed systems